Math 541B HW 5

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Exercise 1. Let $n > m$ be integers. Let A be an $n \times m$ real matrix of known (deterministic) constants. Let $\beta \in \mathbb{R}^m$ be an unknown vector of (deterministic) constants. And let $\varepsilon \in \mathbb{R}^n$ be a random vector with $\mathbb{E}\varepsilon = 0$ and such that ε is a vector of i.i.d. random variables. Define $Y \in \mathbb{R}^n$ by $Y = A\beta + \varepsilon$. Assume that A^TA is invertible. Define $Z := (A^TA)^{-1}A^TY$.

Show that the estimator

$$\left(\frac{1}{n-m}\sum_{i=1}^{n}\big(Y_i - (AZ)_i\big)^2\right)(A^TA)^{-1}$$

is an unbiased estimator of the covariance matrix of $Z := (A^TA)^{-1}A^TY$.

Solution: For a random vector W with finite second moments, its covariance matrix is $\mathrm{Cov}(W) := \mathbb{E}\big[(W - \mathbb{E}W)(W - \mathbb{E}W)^T\big]$, and the expectation of a random matrix is taken entrywise.

The (i,j) entry of $\mathbb{E}[\varepsilon\varepsilon^T]$ is $\mathbb{E}[\varepsilon_i\varepsilon_j]$. If $i \neq j$, then ε_i and ε_j are independent, so $\mathbb{E}[\varepsilon_i\varepsilon_j] = \mathbb{E}[\varepsilon_i]\,\mathbb{E}[\varepsilon_j] = 0$. If $i = j$, then $\mathbb{E}[\varepsilon_i^2] = \sigma^2$ because the ε_i are identically distributed with mean zero. Hence

$$\mathbb{E}[\varepsilon\varepsilon^T] = \sigma^2 I_n.$$

We will repeatedly use the following consequence of linearity of expectation: if M and N are deterministic matrices of compatible sizes, then $\mathbb{E}[M\varepsilon\varepsilon^T N] = M\,\mathbb{E}[\varepsilon\varepsilon^T]\,N$, since each entry of $M\varepsilon\varepsilon^T N$ is a fixed linear combination of the entries of $\varepsilon\varepsilon^T$.

Substituting $Y = A\beta + \varepsilon$,

$$Z = (A^TA)^{-1}A^T(A\beta + \varepsilon) = (A^TA)^{-1}(A^TA)\beta + (A^TA)^{-1}A^T\varepsilon = \beta + (A^TA)^{-1}A^T\varepsilon.$$

Since $\mathbb{E}\varepsilon = 0$, we get $\mathbb{E}Z = \beta$, and therefore $Z - \mathbb{E}Z = (A^TA)^{-1}A^T\varepsilon$. Using Step 1 and $((A^TA)^{-1})^T = (A^TA)^{-1}$,

$$\mathrm{Cov}(Z) = \mathbb{E}\big[(A^TA)^{-1}A^T\varepsilon\varepsilon^TA(A^TA)^{-1}\big] = (A^TA)^{-1}A^T(\sigma^2 I_n)A(A^TA)^{-1} = \sigma^2(A^TA)^{-1}(A^TA)(A^TA)^{-1} = \sigma^2(A^TA$$

Define the $n \times n$ matrix $H := A(A^TA)^{-1}A^T$. It has the following properties.

- H is symmetric: $H^T = A((A^TA)^{-1})^TA^T = A(A^TA)^{-1}A^T = H$.
- H is idempotent: $H^2 = A(A^TA)^{-1}(A^TA)(A^TA)^{-1}A^T = A(A^TA)^{-1}A^T = H$.
- $\mathrm{tr}(H) = m$: by the cyclic property of the trace, $\mathrm{tr}(A(A^TA)^{-1}A^T) = \mathrm{tr}((A^TA)^{-1}A^TA) = \mathrm{tr}(I_m) = m$.
- $HA = A$: indeed $HA = A(A^TA)^{-1}(A^TA) = A$.

Consequently $P := I_n - H$ is symmetric, idempotent (since $P^2 = I_n - 2H + H^2 = I_n - H = P$), satisfies $\operatorname{tr}(P) = n - m$, and satisfies $PA = A - HA = 0$.

Since $AZ = A(A^TA)^{-1}A^TY = HY$, the residual vector is

$$Y - AZ = (I_n - H)Y = P(A\beta + \varepsilon) = PA\beta + P\varepsilon = P\varepsilon.$$

Using that P is symmetric and idempotent,

$$\sum_{i=1}^{n} \big(Y_i - (AZ)_i\big)^2 = \|P\varepsilon\|^2 = \varepsilon^T P^T P\varepsilon = \varepsilon^T P\varepsilon.$$

The quantity $\varepsilon^T P\varepsilon$ is a real number, hence equal to its own trace, and by the cyclic property of the trace,

$$\varepsilon^T P\varepsilon = \operatorname{tr}(\varepsilon^T P\varepsilon) = \operatorname{tr}(P\varepsilon\varepsilon^T).$$

The trace is a linear function of the entries of a matrix, so it commutes with expectation. Using Step 1,

$$\mathbb{E}\left[\sum_{i=1}^{n} \big(Y_i - (AZ)_i\big)^2\right] = \operatorname{tr}\big(P\,\mathbb{E}[\varepsilon\varepsilon^T]\big) = \sigma^2 \operatorname{tr}(P) = \sigma^2(n-m).$$

Since $n > m$, we may divide by $n - m > 0$. Defining

$$\hat{\sigma}^2 := \frac{1}{n-m} \sum_{i=1}^{n} \big(Y_i - (AZ)_i\big)^2,$$

we have shown that $\mathbb{E}[\hat{\sigma}^2] = \sigma^2$.

The matrix $(A^TA)^{-1}$ is deterministic, so by linearity of expectation

$$\mathbb{E}\big[\hat{\sigma}^2 (A^TA)^{-1}\big] = \mathbb{E}[\hat{\sigma}^2]\,(A^TA)^{-1} = \sigma^2 (A^TA)^{-1} = \operatorname{Cov}(Z).$$

Therefore $\hat{\sigma}^2 (A^TA)^{-1}$ is an unbiased estimator of the covariance matrix of Z.

Exercise 2. Assume the one-way ANOVA assumptions. Consider the null hypothesis H_0 that $\beta_1 = \cdots = \beta_p$. Recall that, under this assumption, the F statistic takes the form

$$F = \frac{1}{S^2} \sum_{j=1}^{p} n_i \big(\overline{Y}_j - \overline{Y}\big)^2.$$

The alternative hypothesis H_1 is that $\beta_i \neq \beta_j$ for some $1 \leq i < j \leq p$. We can therefore reject H_0 when F is large.

Show that the generalized likelihood ratio test of H_0 versus H_1 coincides with the hypothesis test we just described. (The likelihood function should just use the Gaussian assumptions for the random variables $Y_1, Y_2, \ldots$.) (Also, you should assume that $\sigma > 0$ is unknown.) (When you form the generalized likelihood ratio, the exponential terms from the Gaussian distribution should eventually become constants.)

Solution: Under the one-way ANOVA assumptions there are $p \geq 2$ groups; group j contains $n_j \geq 1$ observations, and

$$Y_{ij} = \beta_j + \varepsilon_{ij}, \qquad 1 \leq j \leq p, \quad 1 \leq i \leq n_j,$$

where the ε_{ij} are i.i.d. $N(0,\sigma^2)$. The total sample size is $N := \sum_{j=1}^{p} n_j$, and we assume $N > p$. The parameter is $\theta = (\beta_1, \dots, \beta_p, \sigma) \in \Theta := \mathbb{R}^p \times (0,\infty)$, and H_0 corresponds to the subset $\Theta_0 := \{\theta \in \Theta : \beta_1 = \cdots = \beta_p\}$. We write

$$\overline{Y}_j := \frac{1}{n_j} \sum_{i=1}^{n_j} Y_{ij}, \qquad \overline{Y} := \frac{1}{N} \sum_{j=1}^{p} \sum_{i=1}^{n_j} Y_{ij},$$

and define the within-group, between-group and total sums of squares

$$\mathrm{SSW} := \sum_{j=1}^{p} \sum_{i=1}^{n_j} (Y_{ij} - \overline{Y}_j)^2, \qquad \mathrm{SSB} := \sum_{j=1}^{p} n_j (\overline{Y}_j - \overline{Y})^2, \qquad \mathrm{SST} := \sum_{j=1}^{p} \sum_{i=1}^{n_j} (Y_{ij} - \overline{Y})^2.$$

Recall that $S^2 = \mathrm{SSW}/(N-p)$. The index n_i inside the sum defining F refers to the size of the group being summed over, so we write it as n_j; then

$$F = \frac{\mathrm{SSB}}{S^2} = (N-p)\,\frac{\mathrm{SSB}}{\mathrm{SSW}}.$$

(If the F statistic is normalized with an additional factor $1/(p-1)$, it changes only by a positive constant factor, and the argument below is unchanged.)

We note that $\mathrm{SSW} > 0$ with probability one. Indeed, since $N > p$, some group j has $n_j \geq 2$. The random variable $Y_{1j} - Y_{2j}$ is Gaussian with variance $2\sigma^2 > 0$, so $P(Y_{1j} = Y_{2j}) = 0$; and if $Y_{1j} \neq Y_{2j}$, then not all observations in group j equal $\overline{Y}_j$, so $\mathrm{SSW} > 0$. We work on this event of probability one. We will also see below that $\mathrm{SST} \geq \mathrm{SSW}$, so $\mathrm{SST} > 0$ as well.

Since the Y_{ij} are independent with $Y_{ij} \sim N(\beta_j, \sigma^2)$, the likelihood is

$$L(\beta,\sigma) = \prod_{j=1}^{p} \prod_{i=1}^{n_j} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(Y_{ij}-\beta_j)^2}{2\sigma^2}\right) = (2\pi\sigma^2)^{-N/2} \exp\left(-\frac{Q(\beta)}{2\sigma^2}\right),$$

where

$$Q(\beta) := \sum_{j=1}^{p} \sum_{i=1}^{n_j} (Y_{ij} - \beta_j)^2.$$

The generalized likelihood ratio and the corresponding test are

$$\Lambda := \frac{\sup_{\theta \in \Theta_0} L(\theta)}{\sup_{\theta \in \Theta} L(\theta)} \in (0,1], \qquad \text{reject } H_0 \text{ when } \Lambda \leq k,$$

for a constant $k \in (0,1)$ chosen to achieve the desired level. Some texts place $\sup_{\theta \in \Theta \setminus \Theta_0} L(\theta)$ in the denominator instead. This gives the same value: since $p \geq 2$, the set Θ_0 has empty interior in Θ, so $\Theta \setminus \Theta_0$ is dense in Θ, and since L is continuous, its supremum over $\Theta \setminus \Theta_0$ equals its supremum over Θ.

Fix β with $Q(\beta) > 0$ and write $v = \sigma^2 > 0$. Then

$$\ell(v) := \log L = -\frac{N}{2}\log(2\pi) - \frac{N}{2}\log v - \frac{Q(\beta)}{2v}, \qquad \ell'(v) = -\frac{N}{2v} + \frac{Q(\beta)}{2v^2} = \frac{Q(\beta) - Nv}{2v^2}.$$

Thus $\ell'(v) > 0$ for $v < Q(\beta)/N$ and $\ell'(v) < 0$ for $v > Q(\beta)/N$, so ℓ attains its global maximum over $(0, \infty)$ exactly at $v = Q(\beta)/N$. Substituting this value, the exponent becomes $-Q(\beta)/(2Q(\beta)/N) = -N/2$, and

$$\sup_{\sigma>0} L(\beta, \sigma) = \left(\frac{2\pi Q(\beta)}{N}\right)^{-N/2} e^{-N/2}. \tag{1}$$

The exponential term has become the constant $e^{-N/2}$. The right-hand side of (1) is a strictly decreasing function of $Q(\beta)$. Therefore, under each hypothesis, maximizing L over (β, σ) amounts to minimizing $Q(\beta)$ over the allowed β, and then substituting the minimum value into (1).

We minimize $Q(\beta) = \sum_{j=1}^{p} q_j(\beta_j)$, where $q_j(c) := \sum_{i=1}^{n_j}(Y_{ij} - c)^2$. Each q_j depends on a different variable, so it suffices to minimize each q_j separately. The function q_j is a quadratic polynomial in c with $q_j'(c) = -2\sum_i(Y_{ij} - c) = -2n_j(\overline{Y}_j - c)$ and $q_j''(c) = 2n_j > 0$, so its unique global minimizer is $c = \overline{Y}_j$. Hence $\min_\beta Q(\beta) = \mathrm{SSW} > 0$, and by (1),

$$\sup_{\theta\in\Theta} L(\theta) = \left(\frac{2\pi\,\mathrm{SSW}}{N}\right)^{-N/2} e^{-N/2}.$$

Under H_0 we have $\beta_1 = \cdots = \beta_p = \mu$ for some $\mu \in \mathbb{R}$, and $Q = q(\mu) := \sum_j \sum_i (Y_{ij} - \mu)^2$. As before, $q'(\mu) = -2N(\overline{Y} - \mu)$ and $q''(\mu) = 2N > 0$, so the unique minimizer is $\mu = \overline{Y}$ and $\min q = \mathrm{SST}$. By (1),

$$\sup_{\theta\in\Theta_0} L(\theta) = \left(\frac{2\pi\,\mathrm{SST}}{N}\right)^{-N/2} e^{-N/2}.$$

Writing $Y_{ij} - \overline{Y} = (Y_{ij} - \overline{Y}_j) + (\overline{Y}_j - \overline{Y})$ and expanding the square,

$$\mathrm{SST} = \mathrm{SSW} + \sum_{j=1}^{p} n_j(\overline{Y}_j - \overline{Y})^2 + 2\sum_{j=1}^{p}(\overline{Y}_j - \overline{Y})\sum_{i=1}^{n_j}(Y_{ij} - \overline{Y}_j).$$

For each j we have $\sum_{i=1}^{n_j}(Y_{ij} - \overline{Y}_j) = n_j\overline{Y}_j - n_j\overline{Y}_j = 0$, so the last term vanishes and $\mathrm{SST} = \mathrm{SSW} + \mathrm{SSB}$. In particular $\mathrm{SST} \geq \mathrm{SSW} > 0$.

Dividing the results of Steps 3 and 2, the factors $(2\pi/N)^{-N/2}$ and $e^{-N/2}$ cancel, and by Step 4,

$$\Lambda = \left(\frac{\mathrm{SSW}}{\mathrm{SST}}\right)^{N/2} = \left(\frac{\mathrm{SSW}}{\mathrm{SSW} + \mathrm{SSB}}\right)^{N/2} = \left(1 + \frac{\mathrm{SSB}}{\mathrm{SSW}}\right)^{-N/2} = \left(1 + \frac{F}{N-p}\right)^{-N/2}.$$

Define $\phi(u) := \big(1 + u/(N-p)\big)^{-N/2}$ for $u \geq 0$. The function $u \mapsto 1 + u/(N-p)$ is strictly increasing and positive, and $w \mapsto w^{-N/2}$ is strictly decreasing on $(0, \infty)$, so ϕ is strictly decreasing and continuous, with $\phi(0) = 1$ and $\phi(u) \to 0$ as $u \to \infty$. Hence, for every $k \in (0, 1)$,

$$\Lambda \leq k \iff \phi(F) \leq k \iff F \geq c, \qquad \text{where } c := \phi^{-1}(k) = (N-p)\big(k^{-2/N} - 1\big) > 0.$$

Conversely, every threshold $c > 0$ for F arises in this way from $k = \phi(c) \in (0, 1)$. Thus the rejection regions of the generalized likelihood ratio test, $\{\Lambda \leq k\}$, are exactly the rejection regions $\{F \geq c\}$ of the test that rejects H_0 when F is large, with the thresholds related by $c = (N-p)(k^{-2/N} - 1)$. In particular, if c is chosen so that the F test has level α, then the generalized likelihood ratio test with $k = \phi(c)$ has the same rejection region, and hence the same level and the same power function. Therefore the two tests coincide.

Exercise 3. In statistics and other applications, we can be presented with data points $(x_1, y_1), \ldots, (x_n, y_n)$. We would like to find the line $y = mx + b$ which lies "closest" to all of these data points. Such a line is known as a linear regression. There are many ways to define the "closest" such line. The standard method is to use least squares minimization. A line which lies close to all of the data points should make the quantities $(y_i - mx_i - b)$ all very small. We would like to find numbers m, b such that the following quantity is minimized:

$$f(m,b) = \sum_{i=1}^{n} (y_i - mx_i - b)^2.$$

Using the second derivative test, show that the minimum value of f is achieved when

$$m = \frac{\left(\sum_{i=1}^n x_i\right)\left(\sum_{j=1}^n y_j\right) - n\left(\sum_{k=1}^n x_k y_k\right)}{\left(\sum_{i=1}^n x_i\right)^2 - n\sum_{j=1}^n x_j^2} = \frac{\sum_{i=1}^n (x_i - \overline{x})(y_i - \overline{y})}{\sum_{j=1}^n (x_j - \overline{x})^2},$$

$$b = \frac{1}{n}\left(\sum_{i=1}^n y_i - m\sum_{j=1}^n x_j\right) = \overline{y} - m\overline{x}.$$

Briefly explain why this is actually the minimum value of $f(m, b)$. (You are allowed to use the inequality $\left(\sum_{i=1}^n x_i\right)^2 \le n\left(\sum_{i=1}^n x_i^2\right)$.)

Solution: Notation and standing assumption. All sums run from 1 to n. We write $\overline{x} := \frac{1}{n}\sum x_i$ and $\overline{y} := \frac{1}{n}\sum y_i$. We assume that the x_i are not all equal (in particular $n \ge 2$). This assumption is necessary: if $x_1 = \cdots = x_n$, then both denominators in the formula for m vanish, and indeed every line through the point $(x_1, \overline{y})$ minimizes f, so there is no unique minimizer.

A preliminary identity. Expanding the square,

$$\sum (x_j - \overline{x})^2 = \sum x_j^2 - 2\overline{x}\sum x_j + n\overline{x}^2 = \sum x_j^2 - n\overline{x}^2,$$

and multiplying by n gives

$$n\sum x_j^2 - \Big(\sum x_i\Big)^2 = n\sum (x_j - \overline{x})^2. \tag{2}$$

The right-hand side of (2) is nonnegative, which recovers the allowed inequality, and it is zero if and only if $x_j = \overline{x}$ for every j. Under our standing assumption, therefore,

$$\Delta := n\sum x_j^2 - \Big(\sum x_i\Big)^2 > 0. \tag{3}$$

Similarly,

$$\sum (x_i - \overline{x})(y_i - \overline{y}) = \sum x_i y_i - \overline{y}\sum x_i - \overline{x}\sum y_i + n\overline{x}\,\overline{y} = \sum x_i y_i - n\overline{x}\,\overline{y},$$

so that

$$n\sum x_k y_k - \Big(\sum x_i\Big)\Big(\sum y_j\Big) = n\sum (x_i - \overline{x})(y_i - \overline{y}). \tag{4}$$

The function f is a polynomial in (m, b), hence infinitely differentiable on $\mathbb{R}^2$, and its first partial derivatives are

$$f_m(m,b) = -2\sum x_i(y_i - mx_i - b), \qquad f_b(m,b) = -2\sum (y_i - mx_i - b).$$

A point (m, b) is a critical point if and only if both vanish, that is, if and only if

$$m\sum x_i^2 + b\sum x_i = \sum x_i y_i, \tag{5}$$

$$m\sum x_i + nb = \sum y_i. \tag{6}$$

Equation (6) is equivalent to

$$b = \frac{1}{n}\Big(\sum y_i - m\sum x_j\Big) = \overline{y} - m\overline{x}.$$

Substituting this expression into (5) and multiplying by n gives

$$nm\sum x_j^2 + \Big(\sum x_i\Big)\Big(\sum y_j\Big) - m\Big(\sum x_i\Big)^2 = n\sum x_k y_k,$$

that is,

$$m\,\Delta = n\sum x_k y_k - \Big(\sum x_i\Big)\Big(\sum y_j\Big).$$

Since $\Delta \neq 0$ by (3), this has exactly one **Solution:** Multiplying numerator and denominator by -1,

$$m = \frac{(\sum x_i)\,(\sum y_j) - n\sum x_k y_k}{(\sum x_i)^2 - n\sum x_j^2},$$

and by (4) and (2), after cancelling the common factor n,

$$m = \frac{\sum (x_i - \overline{x})(y_i - \overline{y})}{\sum (x_j - \overline{x})^2}.$$

Hence f has exactly one critical point, namely the pair (m, b) given in the statement.

The second partial derivatives are

$$f_{mm} = 2\sum x_i^2, \qquad f_{bb} = 2n, \qquad f_{mb} = f_{bm} = 2\sum x_i,$$

which are constants, independent of (m, b). The discriminant is

$$D = f_{mm}f_{bb} - f_{mb}^2 = 4n\sum x_i^2 - 4\Big(\sum x_i\Big)^2 = 4\Delta > 0$$

by (3), and $f_{bb} = 2n > 0$. By the second derivative test, the critical point (m, b) is a strict local minimum of f.

Step 3: why this is the global minimum. Let (m, b) be the critical point found in Step 1, let $(h, k) \in \mathbb{R}^2$ be arbitrary, and set $r_i := y_i - mx_i - b$. Then

$$f(m+h,\, b+k) = \sum \big(r_i - (hx_i + k)\big)^2 = \sum r_i^2 - 2h\sum x_i r_i - 2k\sum r_i + \sum (hx_i + k)^2.$$

By (5) and (6), $\sum x_i r_i = 0$ and $\sum r_i = 0$, so the two middle terms vanish. Moreover,

$$\sum (hx_i + k)^2 = h^2\sum x_i^2 + 2hk\sum x_i + nk^2 = \frac{1}{2}\big(f_{mm}h^2 + 2f_{mb}hk + f_{bb}k^2\big).$$

Hence, for all $(h, k) \in \mathbb{R}^2$,

$$f(m+h,\, b+k) = f(m, b) + \sum (hx_i + k)^2 \geq f(m, b).$$

Equality holds if and only if $hx_i + k = 0$ for every i. If $h \neq 0$, this would force $x_i = -k/h$ for every i, contradicting the assumption that the x_i are not all equal; hence $h = 0$, and then $k = 0$. Therefore $f(m+h, b+k) > f(m, b)$ for every $(h, k) \neq (0, 0)$, so (m, b) is the unique global minimizer of f, and $f(m, b)$ is the minimum value of f on all of $\mathbb{R}^2$.

In other words: because f is a quadratic polynomial, its second-order Taylor expansion at the critical point is exact, and the quadratic part is given by the constant Hessian, which is positive definite precisely because $D > 0$ and $f_{bb} > 0$. So the local information provided by the second derivative test holds globally.

Exercise 4. Let

$$h(x) := \frac{1}{1 + e^{-x}}, \qquad \forall\, x \in \mathbb{R}.$$

Fix $x \in \mathbb{R}$ and $y \in [0, 1]$. Define $t\colon \mathbb{R}^2 \to \mathbb{R}$ by

$$t(a, b) := \log\Big(\big[h(ax + b)\big]^y \big[1 - h(ax + b)\big]^{1-y}\Big), \qquad \forall\, a, b \in \mathbb{R}.$$

Show that t is concave. Conclude that t has at most one global maximum in the sense that there is some $s \in \mathbb{R}$ such that $t(a, b)$ is maximized at (a, b) only when $ax + b = s$.

Solution: For every $z \in \mathbb{R}$ we have $0 < h(z) < 1$, and

$$1 - h(z) = \frac{e^{-z}}{1 + e^{-z}} = \frac{1}{1 + e^{z}}.$$

In particular $[h(z)]^y[1 - h(z)]^{1-y} > 0$, so t is well defined and real-valued. Moreover

$$\frac{h(z)}{1 - h(z)} = \frac{1 + e^{z}}{1 + e^{-z}} = \frac{e^{z}(e^{-z} + 1)}{1 + e^{-z}} = e^{z}, \qquad \text{so} \qquad \log \frac{h(z)}{1 - h(z)} = z,$$

and $\log(1 - h(z)) = -\log(1 + e^{z})$. Using the rules of logarithms,

$$\log\Big([h(z)]^y[1-h(z)]^{1-y}\Big) = y \log h(z) + (1-y)\log(1-h(z)) = y \log \frac{h(z)}{1 - h(z)} + \log(1-h(z)) = yz - \log(1+e^{z}).$$

Therefore, defining

$$g(z) := yz - \log(1 + e^{z}), \qquad z \in \mathbb{R},$$

we have

$$t(a, b) = g(ax + b) \qquad \text{for all } (a, b) \in \mathbb{R}^2.$$

The function g is infinitely differentiable, and

$$g'(z) = y - \frac{e^{z}}{1 + e^{z}} = y - h(z),$$

where we used $\frac{e^{z}}{1+e^{z}} = \frac{1}{1+e^{-z}} = h(z)$. Differentiating again,

$$g''(z) = -h'(z) = -\frac{e^{-z}}{(1 + e^{-z})^2} = -h(z)\big(1 - h(z)\big).$$

Since $0 < h(z) < 1$, we get $g''(z) < 0$ for every $z \in \mathbb{R}$. A twice differentiable function on $\mathbb{R}$ with everywhere negative second derivative is strictly concave, so g is strictly concave: for all $z_1 \neq z_2$ and $\lambda \in (0,1)$,

$$g\big(\lambda z_1 + (1-\lambda)z_2\big) > \lambda g(z_1) + (1-\lambda)g(z_2).$$

Let $p = (a_1, b_1)$ and $q = (a_2, b_2)$ be points of $\mathbb{R}^2$, let $\lambda \in [0,1]$, and set $z_1 := a_1 x + b_1$ and $z_2 := a_2 x + b_2$. The map $(a,b) \mapsto ax+b$ is linear, so the point $\lambda p + (1-\lambda)q = \big(\lambda a_1 + (1-\lambda)a_2,\ \lambda b_1 + (1-\lambda)b_2\big)$ is mapped to

$$\big(\lambda a_1 + (1-\lambda)a_2\big)x + \lambda b_1 + (1-\lambda)b_2 = \lambda z_1 + (1-\lambda)z_2.$$

Using concavity of g (Step 2),

$$t\big(\lambda p + (1-\lambda)q\big) = g\big(\lambda z_1 + (1-\lambda)z_2\big) \geq \lambda g(z_1) + (1-\lambda)g(z_2) = \lambda t(p) + (1-\lambda)t(q).$$

Hence t is concave on $\mathbb{R}^2$.

The same conclusion can be read off from the Hessian. By the chain rule, $t_a = x\, g'(ax+b)$ and $t_b = g'(ax+b)$, so

$$\nabla^2 t(a,b) = g''(ax+b) \begin{pmatrix} x^2 & x \\ x & 1 \end{pmatrix}.$$

The matrix on the right has trace $x^2 + 1 > 0$ and determinant 0, so its eigenvalues are 0 and x^2+1, and it is positive semidefinite. Since $g''(ax+b) < 0$, the Hessian of t is negative semidefinite at every point, which again shows that t is concave. The determinant of the Hessian is 0 everywhere, and indeed t is not strictly concave: it is constant along each line $\{(a,b) : ax + b = \text{const}\}$. This is why the uniqueness statement must be formulated in terms of the value $ax+b$ rather than the point (a,b).

Suppose that $z_1 \neq z_2$ are both global maximizers of g, and let $M := g(z_1) = g(z_2) = \sup_{\mathbb{R}} g$. By strict concavity,

$$g\left(\frac{z_1 + z_2}{2}\right) > \frac{1}{2}g(z_1) + \frac{1}{2}g(z_2) = M,$$

which contradicts the definition of M. Hence g has at most one global maximizer.

As (a,b) ranges over $\mathbb{R}^2$, the value $ax+b$ ranges over all of $\mathbb{R}$; indeed, already the points $(0,b)$ with $b \in \mathbb{R}$ give every real number. Together with $t(a,b) = g(ax+b)$, this shows that

$$\sup_{(a,b)\in\mathbb{R}^2} t(a,b) = \sup_{z\in\mathbb{R}} g(z).$$

Consequently, (a,b) is a global maximizer of t if and only if $ax+b$ is a global maximizer of g.

If t has no global maximizer, the claim holds trivially (with any $s \in \mathbb{R}$). Otherwise g has a global maximizer, which is unique by Step 4; call it s. Then every global maximizer (a,b) of t satisfies $ax + b = s$, and conversely every (a,b) with $ax + b = s$ is a global maximizer of t. Thus t is maximized at (a,b) only when $ax + b = s$, which is the required statement.